Moon-ki Choi

Postdoctoral Researcher

Materials Research Laboratory, *University of Illinois Urbana-Champaign*

Personal website | Google scholar

Email: choi0652@illinois.edu | Phone: +1 612-229-9732

Education

2018–2023 **Ph.D. in Aerospace Engineering and Mechanics**
University of Minnesota Twin Cities, MN, USA
- Advisor: Prof. Ellad B. Tadmor
- Thesis: Atomistic and continuum simulations of 2D materials with environmental effects

2016–2018 **M.S. in Mechanical Engineering**
Sungkyunkwan University, South Korea
- Advisor: Prof. Moon Ki Kim
- Thesis: Atomic simulation of membrane and its bio-nano organelles

2009–2016 **B.S. in Mechanical Engineering**
Sungkyunkwan University, South Korea

Research Interests

Multiscale & Multiphysics Modeling *Continuum mechanics; Molecular dynamics; Quantum mechanics*
Integrated simulation frameworks that connect quantum, atomistic, and continuum scales for predicting mechanical and electronic properties of materials.

Quantum Materials *2D materials; Topological insulator; Strain and defect engineering*
Atomic-scale defects, large-scale strain fields, and electronic structure of quantum materials.

Machine Learning for Materials *Graph neural networks; Gaussian regression*
Physics-informed machine learning models to accelerate multiscale simulations cosidering uncertainty.

Research Experience

2023–present **Postdoctoral Researcher**
University of Illinois Urbana–Champaign, IL, USA
- Advisor: Prof. Harley T. Johnson

2019–2023 **Research Assistant**
University of Minnesota Twin Cities, MN, USA

2018–2019 **Teaching Assistant**
University of Minnesota Twin Cities, MN, USA

Summer 2017	**Visiting Student**	
	University of Cincinnati, OH, USA	
2016–2018	**Research Assistant**	
	Sungkyunkwan University, South Korea	
	Material Strength & Computational Bioengineering Laboratory	
Fall 2015	**Undergraduate Researcher**	
	Sungkyunkwan University, South Korea	
	Material Strength & Computational Bioengineering Laboratory	

Publications

# indicates co-first authorship.

Under Review/In Revision

[1] S. Frisone, **M. Choi**, J. Hung, J. Essman, Z. You, S. Sinha, A. Birge, H. T. Johnson, M. L. Lee, C. Uher, C. Kurdak, and R. S. Goldman, "**Mechanisms for stress relaxation in BiSb single crystals**", *Submitted (Journal of Applied Physics).*

[2] A. Liu, A. Gil, **M. Choi**, B. A. Hanedar, Z. You, S. Sinha, T. Hakioglu, H. T. Johnson, K. Sun, R. Clarke, C. Uher, C. Kurdak, and R. S. Goldman, "**Probing topological surface states and conduction via extended defects in $(Bi_{1-x}Sb_x)_2Te_3$ films**", *Under review (Applied Physics Letters Materials).*

Preprint

[1] **M. Choi** and E. B. Tadmor, "**Statistical mechanics of a 2d material in a gas reservoir**", *arXiv preprint*, 10.48550/arXiv.2601.16856 (2026).

[2] **M. Choi**, D. Palmer, and H. T. Johnson, "**Graph neural network for unified electronic and interatomic potentials: strain-tunable electronic structures in 2D materials**", *arXiv preprint*, 10.48550/arXiv.2510.16605 (2025).

[3] **M. Choi**, D. Palmer, and H. T. Johnson, "**Interatomic potential development for topological insulator $Bi_{1-x}Sb_x$ and its dislocation by force-following active learning**", *arXiv preprint*, 10.48550/arXiv.2510.18217 (2025).

Published/Accepted

[1] S. Edward, **M. Choi**, and H. T. Johnson, "**Coupling dsmc and akmc to simulate graphite erosion due to hyperthermal oxidation at the nanoscale**", *International Journal of Heat and Mass Transfer* 259, 128317 (2026).

[2] **M. Choi#**, P. Jisuk#, and K. Junpyo, "**Interlocked Woven COFs as Molecular Metamaterials with Snap-Through and Jamming Mechanics**", *Accepted @ Journal of the American Chemical Society* (2026).

[3] S. Sinha, Z. You, S. Smolenski, E. Downey, A. Liu, S. Frisone, A. R. M. Birge, **M. Choi**, A. Shawon, E. Chandler, et al., "**Emerging conduction pathways in semiconducting bismuth-antimony alloys**", *Journal of Physics: Condensed Matter* (2026).

[4] M. T. Ahmed, **M. Choi**, H. T. Johnson, and N. C. Admal, "**Quantifying superlubricity of bilayer graphene from the mobility of interface dislocations**", *ACS Applied Materials & Interfaces* 17, 30197–30211 (2025).

[5] S. Edward, **M. Choi**, and H. T. Johnson, "**Application of time series analysis on kinetic monte carlo simulations of hyperthermal oxidation of graphite**", *The Journal of Physical Chemistry C* 129, 1692–1701 (2025).

[6] M. Kim#, S. Jung#, S. Kim#, **M. Choi#**, J.-H. Hong, K. Kim, C. Park, K. J. Yu, G. D. Cha, S.-H. Sunwoo, Q. Liu, D.-H. Kim, and T.-W. Kim, "**2D silver nanosheet assembly for an isotropic, stretchable, and highly conductive nanomembrane**", *Advanced Materials*, e16002 (2025).

[7] **M. Choi** and H. T. Johnson, "**Mechanics of out-of-plane screw dislocation in a 2D material**", *Extreme Mechanics Letters* 77, 102320 (2025).

[8] S. W. Park, **M. Choi**, B. H. Lee, S. Seo, W. K. Kim, and M. K. Kim, "**An accelerated molecular dynamics study for investigating protein pathways using the bond-boost hyperdynamics method**", *Protein Science* 34, e70073 (2025).

[9] J. R. Gissinger, I. Nikiforov, Y. Afshar, B. Waters, **M. Choi**, D. S. Karls, A. Stukowski, W. Im, H. Heinz, A. Kohlmeyer, and E. B. Tadmor, "**Type label framework for bonded force fields in lammps**", *The Journal of Physical Chemistry B* 128, 3282–3297 (2024).

[10] **M. Choi**, S. H. Sung, R. Hovden, and E. B. Tadmor, "**Elastic plate basis for the deformation and electron diffraction of twisted bilayer graphene on a substrate**", *Physical Review B* 110, 024116 (2024).

[11] **M. Choi#**, M. Pasetto#, Z. Shen#, E. B. Tadmor, and D. Kamensky, "**Atomistically-informed continuum modeling and isogeometric analysis of 2D materials over holey substrates**", *Journal of the Mechanics and Physics of Solids* 170, 105100 (2023).

[12] Y. Zhang#, **M. Choi#**, G. Haugstad, E. B. Tadmor, and D. J. Flannigan, "**Holey substrate-directed strain patterning in bilayer MoS_2**", *ACS Nano* 15, 20253–20260 (2021).

[13] H. Kim, B. H. Lee, **M. Choi**, S. Seo, and M. K. Kim, "**Effects of aquaporin-lipid molar ratio on the permeability of an aquaporin z-phospholipid membrane system**", *PLOS One* 15, e0237789 (2020).

[14] T. Kim, **M. Choi**, H. S. Ahn, J. Rho, H. M. Jeong, and K. Kim, "**Fabrication and characterization of zeolitic imidazolate framework-embedded cellulose acetate membranes for osmotically driven membrane process**", *Scientific Reports* 9, 5779 (2019).

[15] B. H. Lee, S. Jo, **M. Choi**, M. H. Kim, J. B. Choi, and M. K. Kim, "**Normal mode analysis of zika virus**", *Computational Biology and Chemistry* 72, 53–61 (2018).

[16] C. S. Lee#, **M. Choi#**, Y. Y. Hwang, H. Kim, M. K. Kim, and Y. J. Lee, "**Facilitated water transport through graphene oxide membranes functionalized with aquaporin-mimicking peptides**", *Advanced Materials* 30, 1705944 (2018).

[17] **M. Choi**, H. Kim, B. H. Lee, T. Kim, J. Rho, M. K. Kim, and K. Kim, "**Understanding carbon nanotube channel formation in the lipid membrane**", *Nanotechnology* 29, 115702 (2018).

[18] B. H. Lee, S. Seo, M. H. Kim, Y. Kim, S. Jo, **M. Choi**, H. Lee, J. B. Choi, and M. K. Kim, "**Normal mode-guided transition pathway generation in proteins**", *PLOS One* 12, e0185658 (2017).

[19] **M. Choi**, S. Jo, B. H. Lee, M. H. Kim, J. B. Choi, K. Kim, and M. K. Kim, "**Dynamic characteristics of a flagellar motor protein analyzed using an elastic network model**", *Journal of Molecular Graphics and Modelling* 78, 81–87 (2017).

Skills

Computational Language	Python, C++, Fortran90, MATLAB
Quantum Mechanics Simulation	Quantum Espresso, Wannier90, Pythtb, ASE
Classical Atomistic Simulation	LAMMPS, GROMACS, ASE
Continuum Model Simulation	Abaqus, FEniCS
Miscellaneous	Linux, Git, LaTeX, OpenMP

Research Funding Experience

National high-performance computing resource program National Science Foundation (NSF)	***Role: PI***	1,900,000 Service Units ($\approx$ **1,900,000 CPU hours**) awarded	2024–2028
National high-performance computing resource program National Science Foundation (NSF)	***Role: PI***	4,200,000 Service Units ($\approx$ **4,200,000 CPU hours**) awarded	2024–2026
Early Career Research Program National Research Funding, South Korea	***Role: Co-PI***	Proposal preparation and submission	2025
Scholarship and Teaching for Engineering Postdocs Program University of Illinois Urbana–Champaign	***Role: Trainee***	Training in research proposal writing and grant development	2024

Awards and Honors

Best Poster Prize (sponsored by *Journal of Materials Science: Material Theory*, $200)	*Nanomaterials 2025, U.S. National Congress on Computational Mechanics (USACM)*	2025
Travel Award	*Nanomaterials 2025, U.S. National Congress on Computational Mechanics (USACM)*	2025
Doctoral Dissertation Fellowship ($25,000 + tuition)	*University of Minnesota Twin cities*	2023
Outstanding Poster Award	*Korea MultiScale Mechanics Society*	2017
Outstanding Poster Award	*Korean BioChip Society*	2017
Excellence Award for Oral Presentation	*Sungkyunkwan University*	2016

Presentations

[1] T. Ahmed, M. Choi, H. T. Johnson, and N. Admal, "**Quantifying superlubricity of heterostrained bilayer graphene from the mobility of interface dislocations**", *The 11th International Conference on Multiscale Materials Modeling, Prague, Czech Republic*, 2025.

[2] S. Frisone, M. Choi, S. Sinha, Y. Zhang, A. Birge, Z. You, M. L. Lee, C. Uher, H. T. Johnson, C. Kurdak, et al., "**Generating and imaging dislocations in topologically-insulating $Bi_{1-x}Sb_x$ alloys**", *American Physical Society march meeting*, 2025.

[3] A. Gil, A. Liu, M. Choi, H. T. Johnson, R. Clarke, C. Uher, T. Hakioglu, K. Sun, R. Goldman, C. Kurdak, et al., "**Investigating carrier transport through extended defects in $(Bi_{1-x}Sb_x)_2Te_3$ thin films**", *American Physical Society march meeting*, 2025.

[4] T. Hakioglu, B. A. Hanedar, M. Choi, H. T. Johnson, R. Goldman, and C. Kurdak, "**Structural and electronic properties of the $Bi_{1-x}Sb_x$ mono-layers**", *American Physical Society march meeting*, 2025.

[5] M. Choi, D. Palmer, and H. T. Johnson, "**Equivariant graph neural network for unified interatomic potential and tight-binding model**", *The 18th U.S. National Congress on Computational Mechanics, Chicago, IL*, 2025.

[6] M. Choi, D. Palmer, and H. T. Johnson, "**Unified multi-physics graph neural network: deformation-modulated band structures in 2D materials**", *Society of Engineering Science Annual Technical Meeting*, 2025.

[7] M. Choi, D. Palmer, and H. T. Johnson, "**Unified prediction of structural and electronic properties in 2D materials via equivariant graph neural network**", *Nanomaterials 2025, Champaign, IL*, 2025.

[8] M. Choi and E. B. Tadmor, "**Automated comparison of bonded fields and reactive interatomic potentials in OpenKIM**", *Cyberloop Workshop on Nano/Bio-Materials Modeling and Validation*, 2021.

[9] M. Choi, H. Kim, K. Kim, and M. K. Kim, "**Spontaneous insertion of a lipid-coated carbon nanotube into a lipid membrane**", *Multiscale & Multiphysics Mechanics Conference, Ulsan, South Korea*, 2017.

[10] H. Kim, M. Choi, Y. Kim, K. Kim, J. B. Choi, and M. K. Kim, "**A molecular dynamics study about perpendicular nano channel formation of lipid-wrapped cnt**", *Korean Society of Mechanical Engineers Annual Fall Conference, Gangwon-do, South Korea*, 2016.

[11] M. Choi, S. Jo, and M. K. Kim, "**Analysis of rotational switching mechanism of flagellar motor protein**", *12th World Congress on Computational Mechanics, Seoul, South Korea*, 2016.

[12] M. Choi, S. Jo, and M. K. Kim, "**Elastic network model based normal mode analysis for switching mechanism of flagellar motor**", *Korean Society of Mechanical Engineers Annual Spring Conference, Jeju, South Korea*, 2016.

[13] M. Choi, S. Jo, and M. K. Kim, "**Symmetry-constrained normal mode analysis of the bacterial flagellar motor**", *Biophysical Society Annual Meeting, Los Angeles, CA*, 2016.

Teaching Experience

Introduction to Molecular Dynamics Simulation	Invited Lecturer	*Ajou University, South Korea*	2025
Multiscale Methods for Bridging Length and Time Scales (AEM 8551)	Course Material Contributor	*University of Minnesota Twin Cities*	2022
Composite Materials (AEM 4511)	Teaching Assistant	*University of Minnesota Twin Cities*	2019
Mathematical Modeling and Simulation in Aerospace Engineering (AEM 3101)	Teaching Assistant	*University of Minnesota Twin Cities*	2018

Teaching Interests

Solid mechanics, Continuum Mechanics, Machine learning, Computational Mechanics, Elasticity, Multiscale modeling, Molecular dynamics simulation, Modeling Material, Mechanics of Materials, Statics, Dynamics, Numerical Methods.

Research Mentoring

Sharon Edward	University of Illinois Urbana–Champaign	Kinetic Monte Carlo simulation for Thermal Protection System	2023–present
Marios Sotiriou	University of Illinois Urbana–Champaign	Tight-binding model and atomistic simulation for Topological Insulators with defects	2023–present
Georgia Robles	University of Minnesota Twin Cities	Gas reservoir atomistic simulation for Nanoparticle Coalescence	2025–present
Aayushi Bohrey	University of Minnesota Twin Cities	Gas reservoir atomistic simulation for Nanoparticle Coalescence	2025–present
Ji Suk Park	Ajou University, South Korea	Covalent Organic Framework atomistic simulation	2024–present

Service

Symposium Organizer	*18th U.S. National Congress on Computational Mechanics*	2025
Reviewer	*Journal of the Mechanics and Physics of Solids (JMPS), Extreme Mechanics Letters (EML), Computer Methods in Applied Mechanics and Engineering (CMAME), Extreme Mechanics Letters (EML), International Journal of Mechanical Sciences (IJMS)*	2019–present